Problem 3.6

A fund is earning 5% simple interest (see Problem 1.5). Calculate the effective interest rate in the 6th year.

Solution

For simple interest, the accumulated value is given by

$$A(t) = P(1 + it).$$

Here,

$$P = 500, \qquad i = 5\% = 0.05.$$

So,

$$A(t) = 500(1 + it).$$

At time $t = 6$,

$$A(6) = 500\big(1 + 0.05 \cdot 6\big) = 500(1.3) = 650.$$

At time $t = 5$,

$$A(5) = 500\big(1 + 0.05 \cdot 5\big) = 500(1.25) = 625.$$

Therefore, the effective interest rate in the 6th year is

$$i_6 = \frac{A(6) - A(5)}{A(5)} = \frac{650 - 625}{625} = 0.04 = 4\%.$$

Hence,

$$\boxed{i_6 = 4\%.}$$